Diabetic Retinopathy Detection

using Deep Learning

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Dataset: APTOS 2019 Blindness Detection (Kaggle)

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Agenda

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APTOS 2019 – Diabetic Retinopathy, 5 severity grades

02 **Exploratory Data Analysis**
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06 **Experiments & Results**
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07 **Model Comparison**
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08 **Interpretability**
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09 **Conclusions & Future Work**
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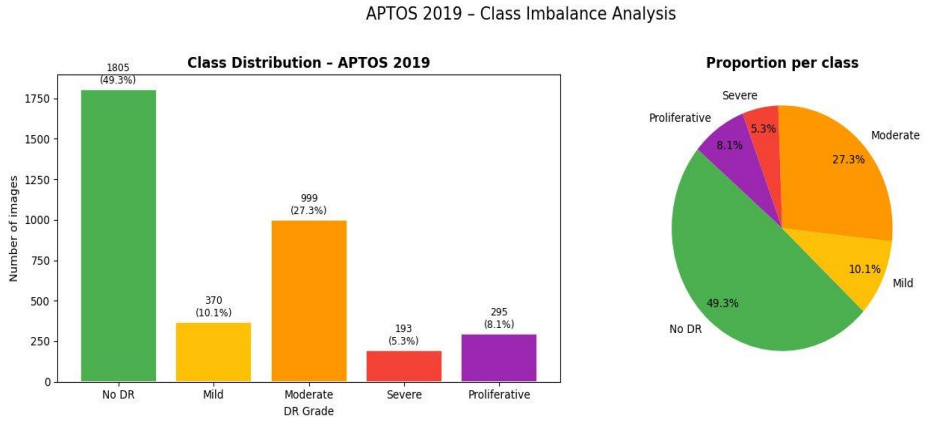
Problem & Dataset – APTOS 2019 Blindness Detection

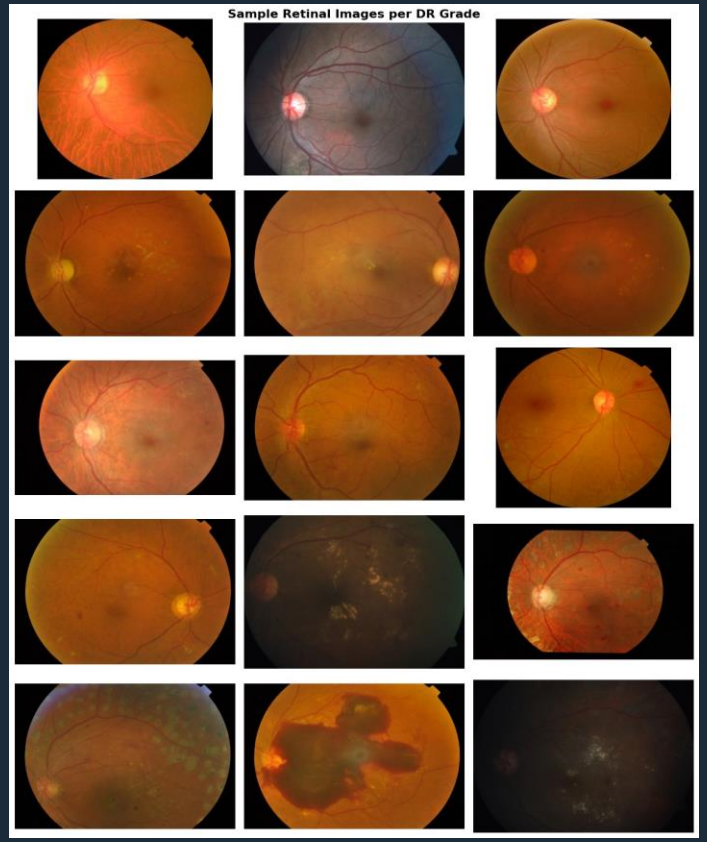
Clinical Context

- Diabetic Retinopathy (DR) is a leading cause of preventable blindness worldwide
- Early detection through retinal fundus photography enables timely treatment
- 5-class ordinal grading: No DR · Mild · Moderate · Severe · Proliferative
- Dataset: 3,662 retinal images from APTOS 2019 Kaggle competition
- Strong class imbalance: No DR = 49.3% vs Severe = 5.3%
- Challenge metric: Quadratic Weighted Kappa (QWK) – penalises ordinal misclassifications



Grade 0 No DR	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Proliferative

Exploratory Data Analysis – Sample Retinal Images per DR Grade



Grade 0 – No DR

Normal retina. Clear vessels, no visible lesions.

Grade 1 – Mild

Microaneurysms only. Earliest detectable change.

Grade 2 – Moderate

More lesions: haemorrhages and hard exudates.

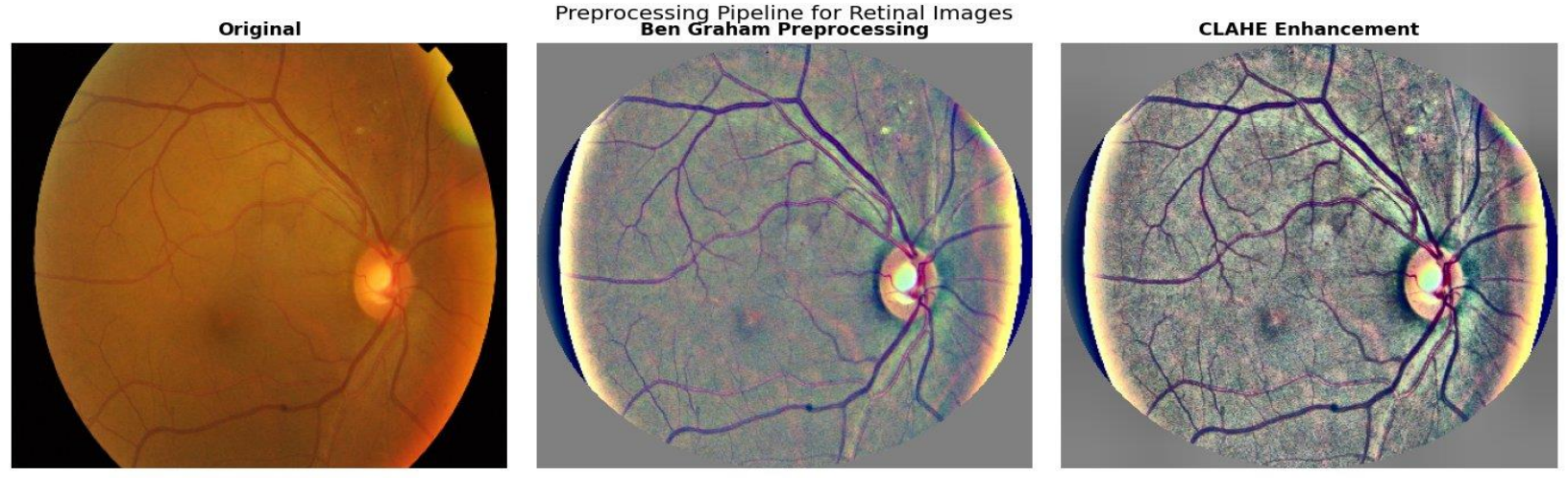
Grade 3 – Severe

Extensive haemorrhages in all 4 quadrants.

Grade 4 – Proliferative

Neovascularisation, vitreous haemorrhage. Sight-threatening.

Preprocessing & Data Augmentation Pipeline



Ben Graham Preprocessing

Circle-crop + Gaussian blur subtraction enhances local contrast of retinal vessels and micro-lesions. Standard technique in APTOS competition winners.

CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalization improves visibility of haemorrhages and exudates in low-contrast regions.

Training Augmentations (Albumentations)

H/V flip · ShiftScaleRotate · RandomBrightness · CoarseDropout · RandomGamma

Class Imbalance Strategy

Inverse-frequency class weights → weighted CrossEntropy + Label Smoothing (ε=0.1) → reduces overconfidence on majority class

Baseline Models – Three Pretrained Architectures (timm)

EfficientNet-B4

Type:	CNN – Compound Scaling
Input:	380 × 380
Weights:	ImageNet-1k
Size:	19M parameters
Exp:	E1 / E4

Fine-grained local features: micro-aneurysms, exudates. Efficient compound scaling of depth/width/resolution.

DenseNet121

Type:	CNN – Dense Connections
Input:	380 × 380
Weights:	ImageNet-1k
Size:	8M parameters
Exp:	E2

Feature reuse via dense skip connections. Strong gradient flow. Good diversity for potential ensemble.

Swin Transformer-S

Type:	Vision Transformer
Input:	224 × 224
Weights:	ImageNet-1k
Size:	50M parameters
Exp:	E3

Hierarchical shifted windows capture global retinal context. Complementary to CNN local features.

All models use a custom classification head: BN → Dropout(0.3) → Linear → Dropout(0.2) → Linear(5 classes)

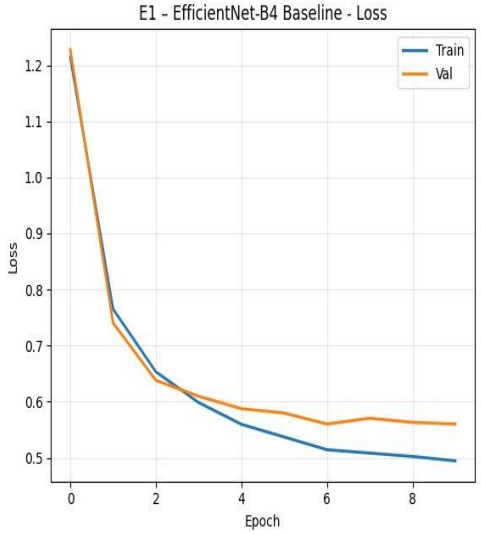
Training Infrastructure & Experiment Design

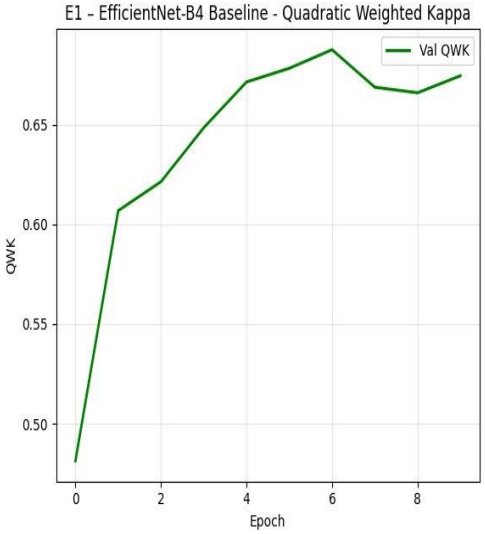
Optimizer	AdamW · lr=1×10^{-4} · weight_decay=1×10^{-4}
Scheduler	CosineAnnealingLR · T_max = n_epochs
Loss	CrossEntropyLoss with class weights + Label Smoothing ε=0.1
Mixed Precision	FP16 via torch.cuda.amp → 2× speedup
Early Stopping	Patience = 3 epochs on val QWK
Mixup	α=0.4 · Applied in E4 full fine-tuning phase
Hardware	Google Colab · NVIDIA T4 GPU · ~14 min/exp

Experiment Overview

Exp	Model	Preproc	Extras
E1	EfficientNet-B4	None	Baseline
E2	DenseNet121	None	Baseline
E3	Swin-S	None	Baseline
E4	EfficientNet-B4	CLAHE + BenGraham	CW + LS + Mixup 2-phase finetune

E1 – EfficientNet-B4 Baseline: Training Results



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RESUMEN ENTRENAMIENTO
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Best QWK: 0.6875
Best Epoch: 7
Final Val Loss: 0.5600
Total Epochs: 10

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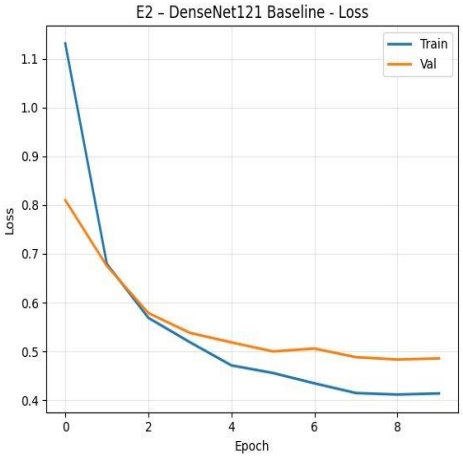
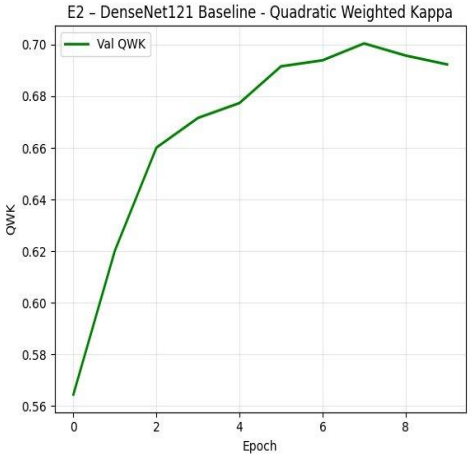
Best QWK	**0.6875**
Best Epoch	**7 / 10**
Val Loss	**0.5600**
Test Acc	**80.0%**
Test QWK	**0.8479**
Test AUC	**0.9347**

Observation: Steady QWK improvement to epoch 7, then early stopping. Train/val loss well aligned – no overfitting. High test QWK (0.848) despite simple baseline setup.

E2 – DenseNet121 Baseline: Training Results



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RESUMEN ENTRENAMIENTO
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Best QWK: 0.7004
Best Epoch: 8
Final Val Loss: 0.4855
Total Epochs: 10

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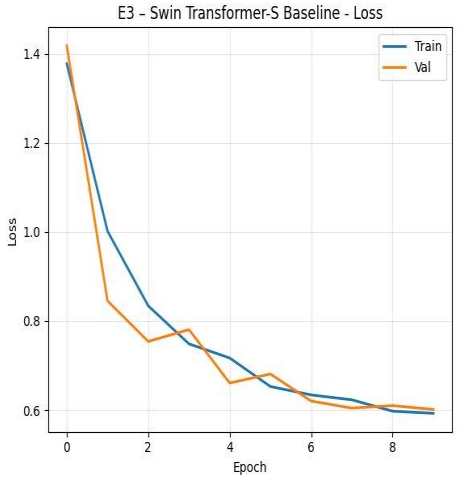
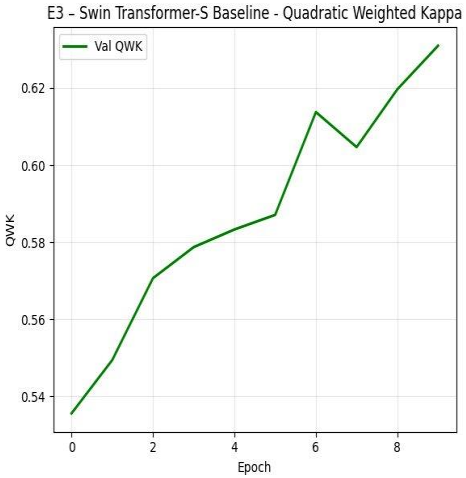
Best QWK	**0.7004**
Best Epoch	**8 / 10**
Val Loss	**0.4855**
Test Acc	**82.7%**
Test QWK	**0.8820**
Test AUC	**0.9474**

★ **BEST MODEL**

Observation: DenseNet121 achieves the highest test QWK (0.882) and AUC (0.947) among all models. Dense connections provide strong feature reuse. Lower val loss (0.485) indicates better calibration.

E3 – Swin Transformer-S Baseline: Training Results



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RESUMEN ENTRENAMIENTO
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Best QWK: 0.6310
Best Epoch: 10
Final Val Loss: 0.6014
Total Epochs: 10

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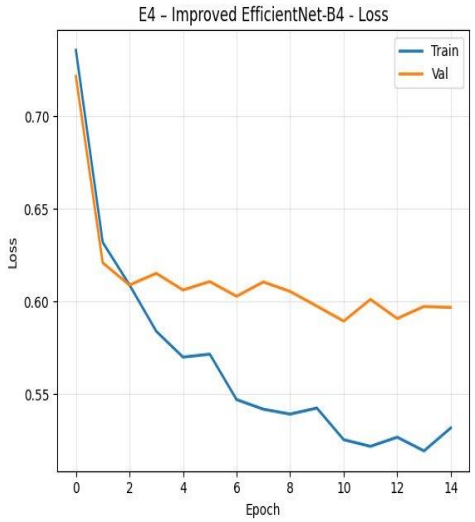
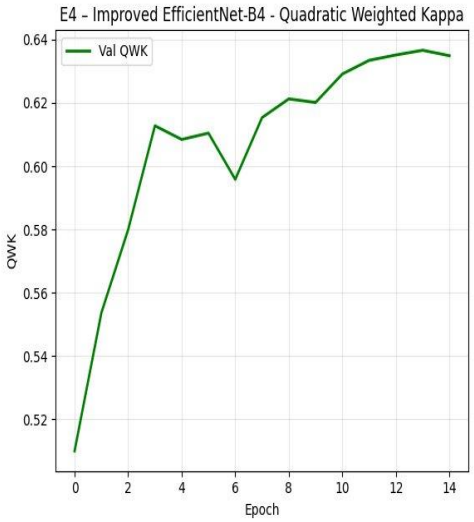
Best QWK	**0.6310**
Best Epoch	**10 / 10**
Val Loss	**0.6014**
Test Acc	**77.4%**
Test QWK	**0.8346**
Test AUC	**0.9397**

Observation: Swin Transformer still converging at epoch 10 – needs more epochs. Lower accuracy reflects 224×224 input vs 380×380 for CNNs. High AUC (0.940) shows good discriminative ability despite lower accuracy.

E4 – Improved EfficientNet-B4: CLAHE + BenGraham + Mixup



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RESUMEN ENTRENAMIENTO
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Best QWK: 0.6366
Best Epoch: 14
Final Val Loss: 0.5968
Total Epochs: 15

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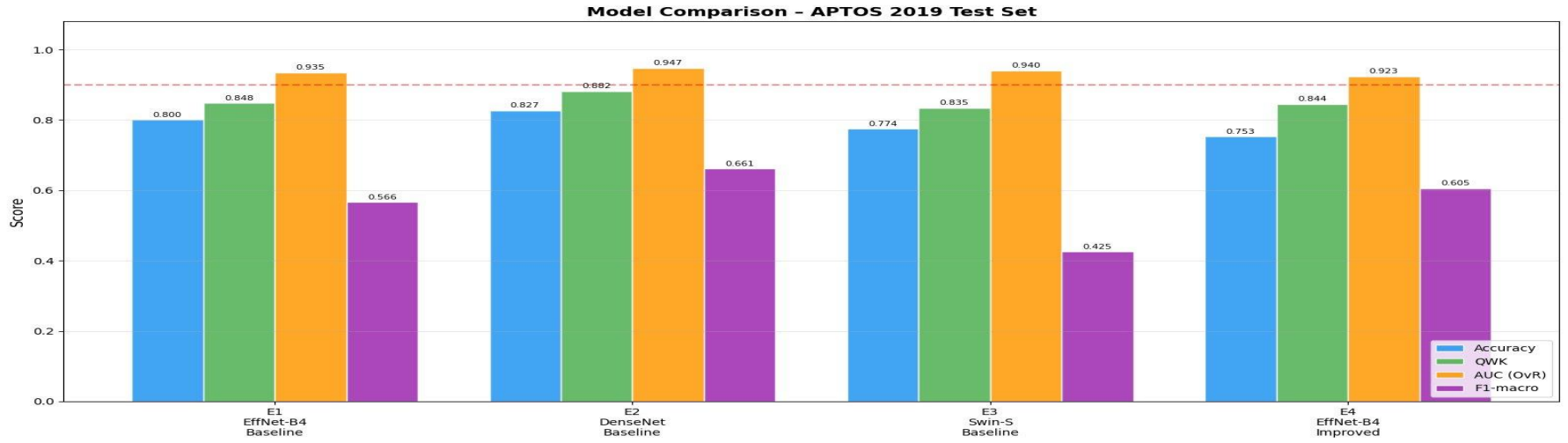
Best QWK	**0.6366**
Best Epoch	**14 / 15**
Val Loss	**0.5968**
Test Acc	**75.3%**
Test QWK	**0.8444**
Test AUC	**0.9230**

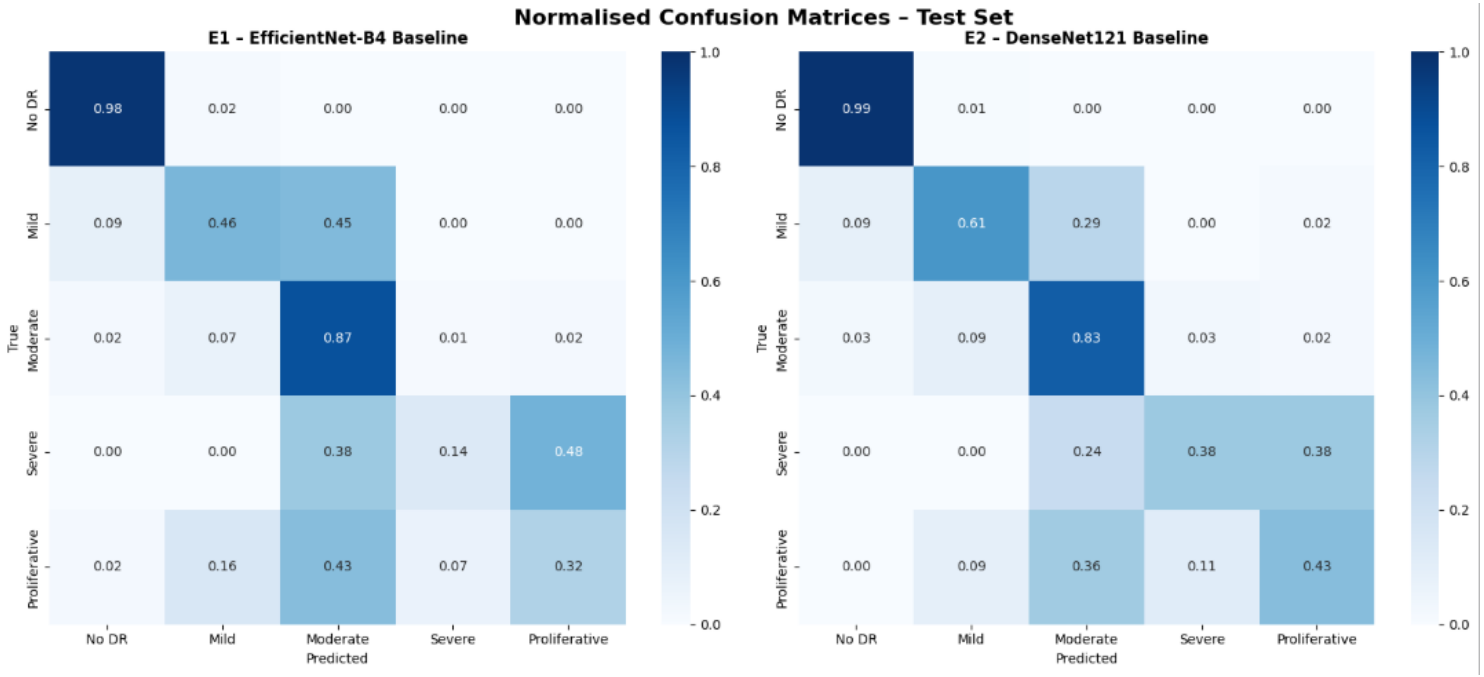
Observation: 2-phase training (head-only → full fine-tune) with Mixup regularization. Steady val loss. Best val epoch = 14 of 15 – more epochs could improve further. Comparable test QWK to E1 baseline.

Model Comparison – APTOS 2019 Test Set



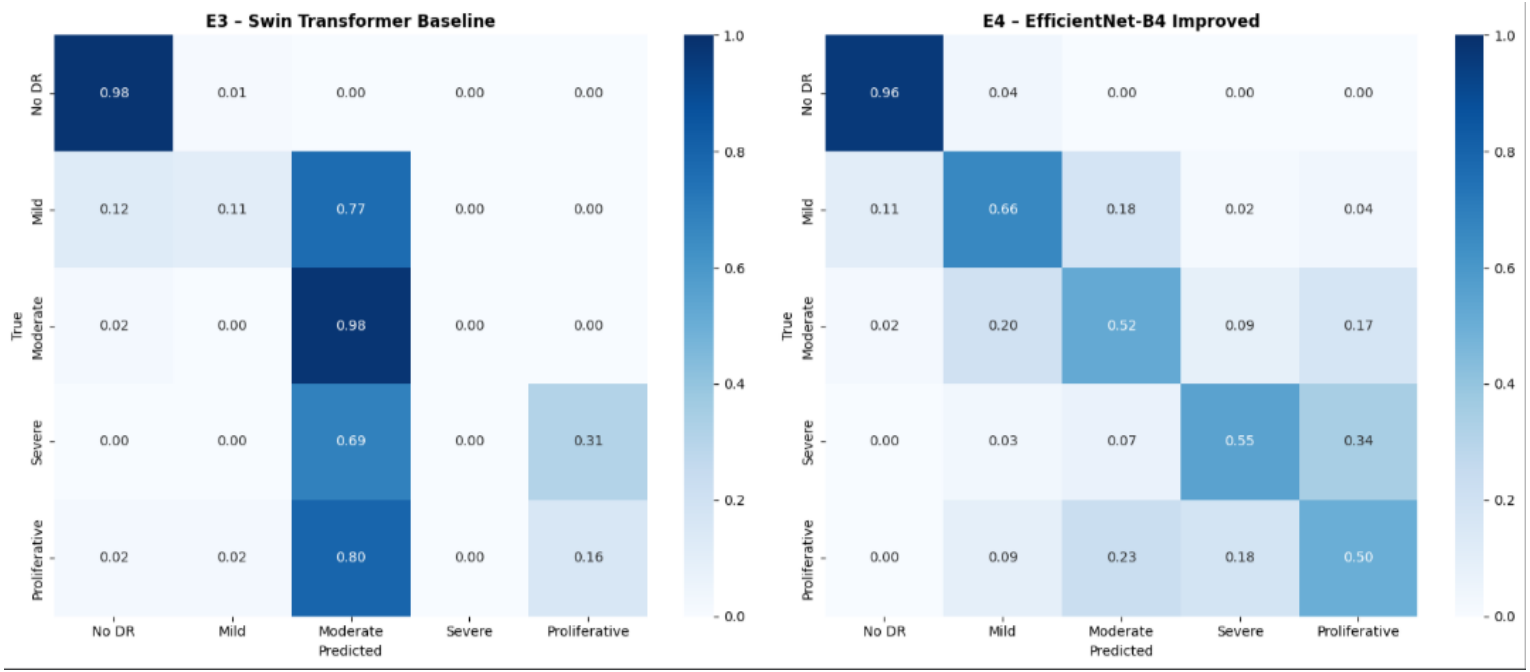
Experiment	Accuracy	QWK ★	AUC (OvR)	F1-macro
E1 – EfficientNet-B4 Baseline	0.8000	0.8479	0.9347	0.5658
E2 – DenseNet121 Baseline	**0.8273**	**0.8820 ★**	**0.9474 ★**	**0.6612 ★**
E3 – Swin-S Baseline	0.7745	0.8346	0.9397	0.4251
E4 – EfficientNet-B4 Improved	0.7527	0.8444	0.9230	0.6052

Normalised Confusion Matrices – All Models

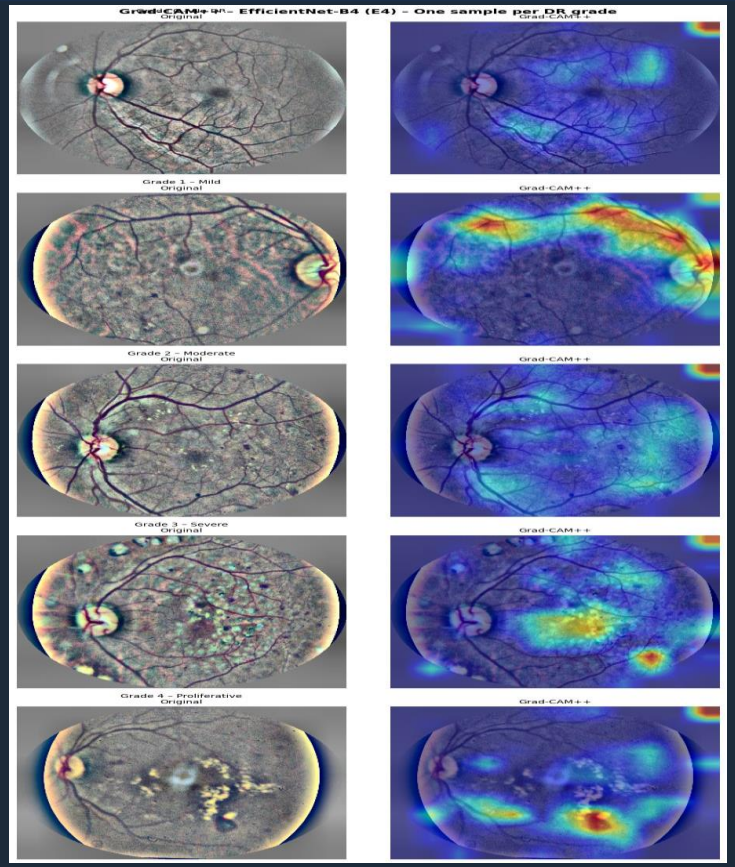


Key finding: All models excel on No DR (Grade 0) due to class abundance. Minority classes (Severe, Proliferative) remain challenging – best handled by E2 (DenseNet121). E3 (Swin) shows collapse toward Moderate class.

Normalised Confusion Matrices – All Models



Interpretability – Grad-CAM++ Visualisation (E4 – EfficientNet-B4)



Grade 0 – No DR

Attention distributed broadly. No focal activation. Model confirms absence of lesions.

Grade 1 – Mild

Activation near optic disc region where micro-aneurysms typically first appear.

Grade 2 – Moderate

Stronger, more diffuse activation. Haemorrhage and exudate regions highlighted.

Grade 3 – Severe

High activation across all quadrants – consistent with 4-quadrant haemorrhage criterion.

Grade 4 – Proliferative

Concentrated hotspots on neovascularisation and vitreous haemorrhage areas.

Final Results Summary – APTOS 2019 Test Set

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FINAL RESULTS SUMMARY – APTOS 2019 Test Set
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                        Experiment Accuracy    QWK AUC (OvR) F1-macro
E1 – EfficientNet-B4 (Baseline)   0.8000 0.8479    0.9347   0.5658
    E2 – DenseNet121 (Baseline)   0.8273 0.8820    0.9474   0.6612
         E3 – Swin-S (Baseline)   0.7745 0.8346    0.9397   0.4251
E4 – EfficientNet-B4 (Improved)   0.7527 0.8444    0.9230   0.6052
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Best QWK

0.8820

E2 DenseNet121

Best AUC

0.9474

E2 DenseNet121

Best Accuracy

82.7%

E2 DenseNet121

Best F1-macro

0.6612

E2 DenseNet121

Key Takeaways

- DenseNet121 outperformed all models including the improved EfficientNet-B4 variant
- QWK >> Accuracy as the relevant metric – ordinal nature of grading must be respected
- Class imbalance remains the primary challenge: Severe & Proliferative grades hardest to classify
- CLAHE + Ben Graham preprocessing enhanced vessel visibility but did not always improve final metrics
- Grad-CAM++ confirms clinically meaningful attention – model focuses on lesion-rich regions

Conclusions & Future Work

Conclusions

- Deep learning achieves strong DR grading performance on APTOS 2019 (best QWK = 0.882)
- DenseNet121 with dense skip-connections excels at feature reuse for medical imaging
- QWK is the appropriate metric for ordinal classification – accuracy alone is misleading
- Advanced preprocessing (CLAHE, BenGraham) improves interpretability; gain depends on architecture
- Grad-CAM++ confirms model attends to clinically relevant retinal structures per DR grade
- All models trained in <20 min on T4 GPU using FP16 mixed precision

Future Work

- Test-Time Augmentation (TTA) – average predictions over augmented versions to boost QWK
- Ensemble of E1+E2+E3 – soft-voting across architectures for complementary strengths
- External validation on Messidor-2 or EyePACS datasets to assess generalization
- Lesion-level detection with YOLO for explainable region proposals
- Ordinal regression head instead of softmax – encode grade ordering explicitly
- Larger resolution (512×512) and longer training with progressive resizing

Thank You

Questions & Discussion

Luis Carlos Velandia Cuevas · Juan Esteban Velez · Juan Cumbe

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0.882	0.947	4	3,662
Best QWK	Best AUC	Experiments	Images